Assignment

An assignment for class **MET CS 566** at Boston University, instructed by **Ming Zhang**. Student Zhang, Jin Yu submitted on **October 2, 2022**.

Question 1

Use Strassen's algorithm to compute the following matrix product. Show your work in different computation steps:

$$C = \begin{pmatrix} 1 & 3 \\ 7 & 5 \end{pmatrix} \begin{pmatrix} 6 & 8 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 18 & 14 \\ 62 & 66 \end{pmatrix}$$

Let C be the multiplication product, A be the first matrix and B be the second matrix:

$$C = A \times B$$

We can then name each value:

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

Let's define the following 7 matrices:

$$\begin{aligned} M_1 &= (A_{11} + A_{22}) \times (B_{11} + B_{22}) \\ M_2 &= (A_{21} + A_{22}) \times B_{11} \\ M_3 &= A_{11} \times (B_{12} - B_{22}) \\ M_4 &= A_{22} \times (B_{21} - B_{11}) \\ M_5 &= (A_{11} + A_{12}) \times B_{22} \\ M_6 &= (A_{21} - A_{11}) \times (B_{11} + B_{12}) \\ M_7 &= (A_{12} - A_{22}) \times (B_{21} + B_{22}) \end{aligned}$$

Put A and B in the matrices:

$$\begin{aligned} M_1 &= (1 + 5) \times (6 + 2) = 48 \\ M_2 &= (7 + 5) \times 6 = 72 \\ M_3 &= 1 \times (8 - 2) = 6 \\ M_4 &= 5 \times (4 - 6) = -10 \\ M_5 &= (1 + 3) \times 2 = 8 \\ M_6 &= (7 - 1) \times (6 + 8) = 84 \\ M_7 &= (3 - 5) \times (4 + 2) = -12 \end{aligned}$$

We can use the matrices in the following formula to calculate each value in the multiplication product's matrix:

$$C_{11} = M_1 + M_4 - M_5 + M_7 = 48 + (-10) - 8 + (-12) = 18$$
$$C_{12} = M_3 + M_5 = 6 + 8 = 14$$
$$C_{21} = M_2 + M_4 = 72 + (-10) = 62$$
$$C_{22} = M_1 + M_3 - M_2 + M_6 = 48 + 6 - 72 + 84 = 66$$

We can verify the answer with Matrix Multiplication:

$$(1 \times 6) + (3 \times 4) = 18$$
$$(1 \times 8) + (3 \times 2) = 14$$
$$(7 \times 6) + (5 \times 4) = 62$$
$$(7 \times 8) + (5 \times 2) = 66$$

Question 2

Write pseudocode for Strassen's Algorithm (assume the input is the power of 2). You can use similar pseudocode syntax similar to the CLRS book or our lecture slides.

```
Strassen's Algorithm
Parameter: A, B (Matrix, 2 level array)

l = A.row.length
C = l x l matrix
if l == 1
    C[1][1] = A[1][1] * B[1][1]
else
    ATL = A[1..l / 2][1..l / 2]
    ATR = A[1..l / 2][l / 2 + 1..l]
    ABL = A[l / 2 + 1..l][1..l / 2]
    ABR = A[l / 2 + 1..l][l / 2 + 1..l]
    BTL = B[1..l / 2][1..l / 2]
    BTR = B[1..l / 2][l / 2 + 1..l]
    BBL = B[l / 2 + 1..l][1..l / 2]
    BBR = B[l / 2 + 1..l][l / 2 + 1..l]

    # The followings are sum of matrices. S1, ..., S0 are matrices
    S1 = BTR - BBR
    S2 = ATL + ATR
    S3 = ABL + ABR
    S4 = BBL - BTL
    S5 = ATL + ABR
    S6 = BTL + BBR
    S7 = ATR - ABR
    S8 = BBL + BBR
    S9 = ABL - ATL
    S0 = BTL + BTR

    # Recursively divide into smaller matrices
    M1 = Strassen's Algorithm(S5, S6)
    M2 = Strassen's Algorithm(S3, BTL)
    M3 = Strassen's Algorithm(ATL, S1)
    M4 = Strassen's Algorithm(ABR, S4)
    M5 = Strassen's Algorithm(S2,  BBR)
    M6 = Strassen's Algorithm(S9,  S0)
    M7 = Strassen's Algorithm(S7, S8)

    C[1..l / 2][1..l / 2] = M1 + M4 - M5 + M7
    C[1..l / 2][l / 2 + 1..l] = M3 + M5
    C[l / 2 + 1..l][1..l / 2] = M2 + M4
    C[l / 2 + 1..l][l / 2 + 1..l] = M1 + M3 - M2 - M6
    return C
```

Question 3

Is it possible to use Strassen's Algorithm to compute the following matrix multiplication?

$$C = \begin{pmatrix} 1 & 3 & 2 \\ 7 & 5 & 2 \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 6 & 8 & 6 \\ 4 & 2 & 1 \\ 5 & 3 & 1 \end{pmatrix}$$

First of all, we know strassen's algorithm requires matrices with $2^n \times 2^n$:

$$\begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} & B_{13} & B_{14} \\ B_{21} & B_{22} & B_{23} & B_{24} \\ B_{31} & B_{32} & B_{33} & B_{34} \\ B_{41} & B_{42} & B_{43} & B_{44} \end{pmatrix}$$

So we pad the matrices with zeroes:

$$\begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 7 & 5 & 2 & 0 \\ 4 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 6 & 8 & 6 & 0 \\ 4 & 2 & 1 & 0 \\ 5 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

After padding with zeroes, the inputs are eligible for Strassen's algorithm and we can calculate it with steps mentioned in Question 1 and Question 2.

Since we can pad zeroes to a 3×3 matrix, we should be able to use Strassen's algorithm for **any** matrix multiplication as long as we can pad zeroes the closet 2^n dimension. For example, a 7×7 matrix can be a 8×8 matrix by padding a column and a row with zeroes.

Padding zeroes **doesn't result in any increasing running time** because the length of padding columns/rows will not be larger than $2 \times length\ of\ rows$.

Question 4

Use the Substitution Method for solving the following recurrences:

$T(n) = T(n-1) + n$ is $\mathcal{O}(n^2)$

We need to prove $T(n) \leq \mathcal{O}(n^2)$ for a constant $c > 0$. We assume this bond should hold for all positive $m < n$, in particular for $m = n - 1$, $T(n-1) \leq c \times (n-1)^2$.

$$\begin{cases} T(n) = T(n-1) + n \\ T(n-1) \leq c \times (n-1)^2 \end{cases}$$

$$\begin{gathered} T(n) \leq (c \times (n-1)^2 - 1) + n \\ T(n) \leq (c \times (n^2 - (2 \times n)) + 1 - 1) + n \\ T(n) \leq (c \times (n^2 - (2 \times n))) + n \\ T(n) \leq (c \times n^2) - (c \times 2 \times n) + n \end{gathered}$$

$\leq c \times n^2$ holds as long as $c \geq 1$.

$T(n) = T(\lceil \frac{n}{2} \rceil) + 1$ is $\mathcal{O}(\log^n)$

We need to prove $T(n) \leq \mathcal{O}(\log^n) - d$ for constants $c > 0$ and $d \geq 0$. We assume this bond should hold for all positive $m < n$, in particular for $m = \lceil \frac{n}{2} \rceil$, $T(\lceil \frac{n}{2} \rceil) \leq (c \times \log^{\lceil \frac{n}{2} \rceil}) - d$.

$$\begin{cases} T(n) = T(\lceil \frac{n}{2} \rceil) + 1 \\ T(\lceil \frac{n}{2} \rceil) \leq (c \times \log^{\lceil \frac{n}{2} \rceil}) - d \end{cases}$$

$$\begin{gathered} T(n) \leq ((c \times \log^{\lceil \frac{n}{2} \rceil}) - d) + 1 \\ T(n) \leq c \times \log^{\lceil \frac{n}{2} \rceil} \end{gathered}$$

$\leq (c \times \log^{\lceil \frac{n}{2} \rceil}) - d$ holds as long as $c \geq 1$ and $d \geq 0$.

Question 5

Use the Master Method for solving the following recurrences:

$$T(n) = a \times T \times (\frac{n}{b}) + \mathcal{O}(n^d)$$

$$T(n) = \begin{cases} \mathcal{O}(n^d \times \log^n) \; if \; a = b^d \\ \mathcal{O}(n^d) \; if \; a < b^d \\ \mathcal{O}(n^{\log_b^a}) \; if \; a > b^d \end{cases}$$

$T(n) = 2T(\frac{n}{4}) + 1$

$$a = 2, b = 4, d = 0. \; a > b^d, \mathcal{O}(n^{\log_4^2})$$

$T(n) = 2T(\frac{n}{4}) + \sqrt{n}$

$$a = 2, b = 4, d = \frac{1}{2}. \; a = b^d, \mathcal{O}(\sqrt{n} \times \log^n)$$

$T(n) = 2T(\frac{n}{4}) + n$

$$a = 2, b = 4, d = 1. \; a < b^d, \mathcal{O}(n)$$